

Lancaster University College
at Beijing Jiaotong University

Digital Signal Processing

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All pass filter and minimum phase system

- All pass filter and its characteristics
- Definition of Minimum Phase System
- Application of All Pass Filter
- Application of Minimum Phase System

Why discuss universal and minimum phase systems?

- Phase equalization can be achieved using an all pass filter
- The minimum phase system can be used for amplitude equalization
- These two systems can be used to represent any causal system

All pass digital filter

➢ Definition:

$A_m(z)$ represents the system function of an m-order real coefficient all pass filter

$$A_m(z) = \frac{d_m + d_{m-1}z^{-1} + \cdots + d_1 z^{-(m-1)} + z^{-m}}{1 + d_1 z^{-1} + \cdots + d_{m-1}z^{-(m-1)} + d_m z^{-m}}$$

$$\left|A_m(\mathrm{e}^{\mathrm{j}\Omega})\right| = 1$$

All pass digital filter

➢A first-order causally stable complex coefficient all pass digital filter

$$A_1(z)=\frac{z^{-1}-d^*}{1-dz^{-1}} \qquad |d|<1 \qquad d=re^{j\theta}$$

(a) amplitude response

$$A_1(e^{j\Omega})=\frac{e^{-j\Omega}-d^*}{1-de^{-j\Omega}}=e^{-j\Omega}\frac{1-d^*e^{j\Omega}}{1-de^{-j\Omega}}=e^{-j\Omega}\frac{1-re^{-j\theta}e^{j\Omega}}{1-re^{j\theta}e^{-j\Omega}}$$

The numerator and denominator are conjugate with each other

$$\left|A_1(e^{j\Omega})\right|=1$$

All pass digital filter

- A first-order causally stable complex coefficient all pass digital filter

$$A_1(z)=\frac{z^{-1}-d^*}{1-dz^{-1}}=z^{-1}\frac{1-d^*z}{1-dz^{-1}} \qquad |d|<1 \qquad d=r\mathrm{e}^{\mathrm{j}\theta}$$

(b) phase response

$$A_1(\mathrm{e}^{\mathrm{j}\Omega})=\mathrm{e}^{-\mathrm{j}\Omega}\frac{1-r\mathrm{e}^{-\mathrm{j}\theta}\mathrm{e}^{\mathrm{j}\Omega}}{1-r\mathrm{e}^{\mathrm{j}\theta}\mathrm{e}^{-\mathrm{j}\Omega}}=\mathrm{e}^{-\mathrm{j}\Omega}\frac{1-r\cos(\Omega-\theta)-\mathrm{j}\cdot r\sin(\Omega-\theta)}{1-r\cos(\Omega-\theta)+\mathrm{j}\cdot r\sin(\Omega-\theta)}$$

$$\varphi(\Omega)=-\Omega-2\arctan\frac{r\sin(\Omega-\theta)}{1-r\cos(\Omega-\theta)}$$

All pass digital filter

(b) phase response

$$\varphi(\Omega) = -\Omega - 2\arctan\frac{r\sin(\Omega-\theta)}{1-r\cos(\Omega-\theta)}$$

$$\frac{\mathrm{d}\varphi(\Omega)}{\mathrm{d}\Omega} = -\frac{1-r^2}{(1-r\cos(\Omega-\theta))^2 + r^2\sin^2(\Omega-\theta)} < 0$$

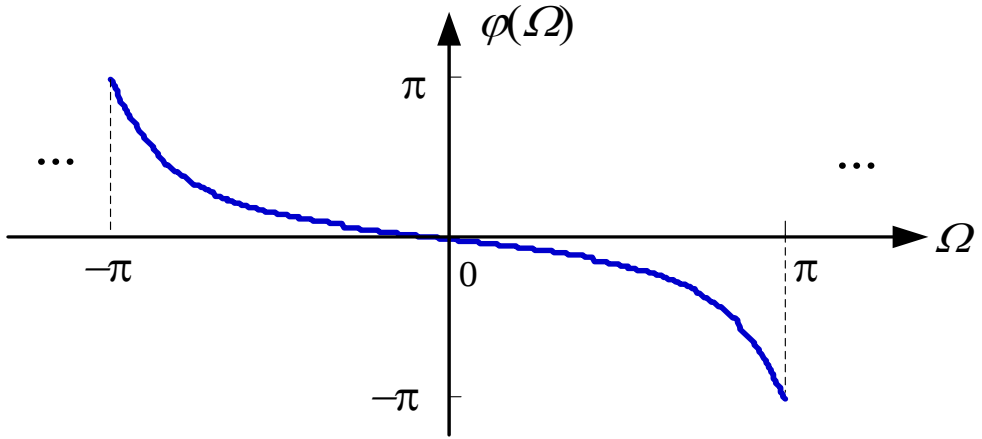
The phase response of a first-order all pass digital filter decreases monotonically.

All pass digital filter

Phase response of first-order all pass digital filter

$$\varphi(\Omega) = -\Omega - 2\arctan\frac{r\sin(\Omega-\theta)}{1-r\cos(\Omega-\theta)}$$

$$d = re^{j\theta}$$



Monotonically decreasing

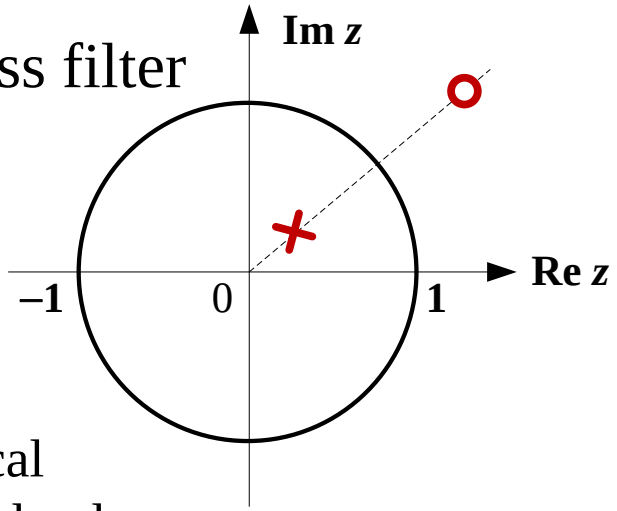
All pass digital filter

➢First-order complex coefficient

$$A_1(z)=\frac{z^{-1}-d^*}{1-dz^{-1}}=\frac{1-d^*z}{z-d}$$

$$|d|<1 \quad d=re^{j\theta}$$

(c) The poles and zeros of a first-order all pass filter



pole: $p_1=d=re^{j\theta}$

Zero: $z_1=1/d^*=(1/r)e^{j\theta}$

$p_1=(\frac{1}{z_1})^*$ There is a conjugate reciprocal relationship between zero and pole

All pass digital filter

- M-order real coefficient all pass filter

$$A_m(z)=\frac{d_m+d_{m-1}z^{-1}+\cdots+d_1z^{-(m-1)}+z^{-m}}{1+d_1z^{-1}+\cdots+d_{m-1}z^{-(m-1)}+d_mz^{-m}}=\frac{z^{-m}D_m(z^{-1})}{D_m(z)}$$

(a) Amplitude response

$$A_m(z)A_m(z^{-1})=\frac{z^{-m}D_m(z^{-1})}{D_m(z)}\frac{z^{m}D_m(z)}{D_m(z^{-1})}=1$$

By utilizing the symmetry of real sequence DTFT, there are

$$\left|A_m(\mathrm{e}^{\mathrm{j}\Omega})\right|^2=A_m(z)A_m(z^{-1})\Big|_{z=\mathrm{e}^{\mathrm{j}\Omega}}=1$$

All pass digital filter

(b) The poles and zeros of *m*-order real coefficient all pass filters

$$A_m(z) = \frac{z^{-m} D_m(z^{-1})}{D_m(z)} = K \frac{(z - z_1)(z - z_2)\cdots(z - z_m)}{(z - p_1)(z - p_2)\cdots(z - p_m)}$$

$$= K \frac{\prod_{i=1}^{m}(z - z_i)}{\prod_{i=1}^{m}(z - p_i)}$$

First order All Pass Filter

$$p_i = \left(\frac{1}{z_i}\right)^*$$

The *m*-order real coefficient all pass system can be decomposed into a cascade of m first-order all pass systems.

All pass digital filter

(c) Phase response of *m*-order real coefficient all pass filter

$$A_m(z)=\frac{z^{-m}D_m(z^{-1})}{D_m(z)}=K\frac{(z-z_1)(z-z_2)\cdots(z-z_m)}{(z-p_1)(z-p_2)\cdots(z-p_m)} \qquad p_i=(\frac{1}{z_i})^*$$

$$\varphi(\Omega)=\varphi_1(\Omega)+\varphi_2(\Omega)+\cdots+\varphi_m(\Omega)$$

due to $A_m(e^{j0})=1$ so $\varphi(0)=0$

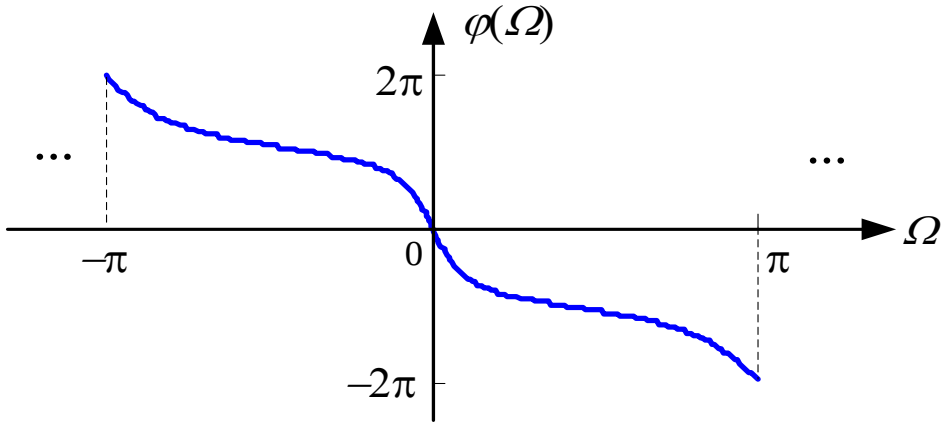
When Ω changes from 0 to π $\varphi_1(\Omega)<0,\varphi_2(\Omega)<0,\cdots,\varphi_m(\Omega)<0$

So for an *m*-order real coefficient all pass system, its phase response is non positive and decreasing as Ω changes from 0 to π.

All pass digital filter

(c)Phase response of *m*-order real coefficient all pass filter

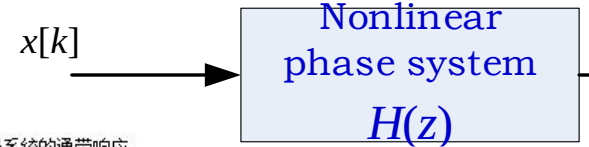
$$\varphi(\Omega)=\varphi_1(\Omega)+\varphi_2(\Omega)+\cdots+\varphi_m(\Omega)$$

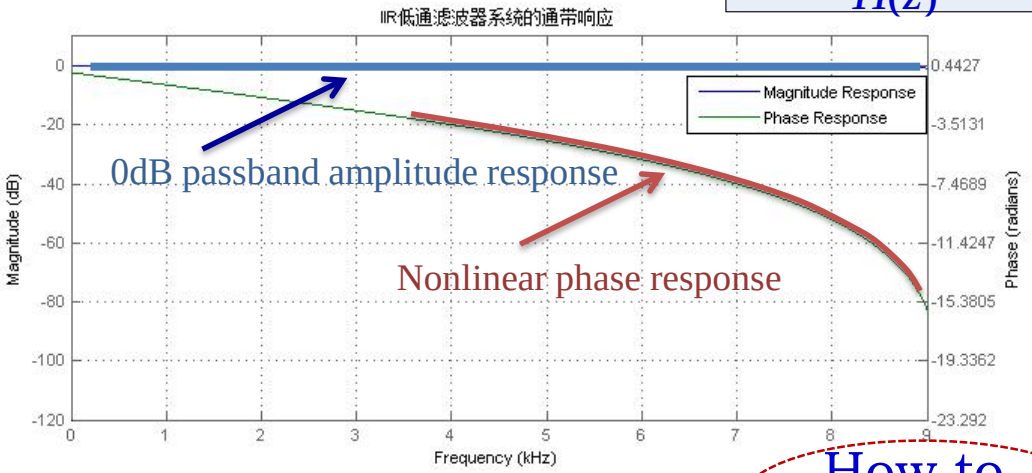


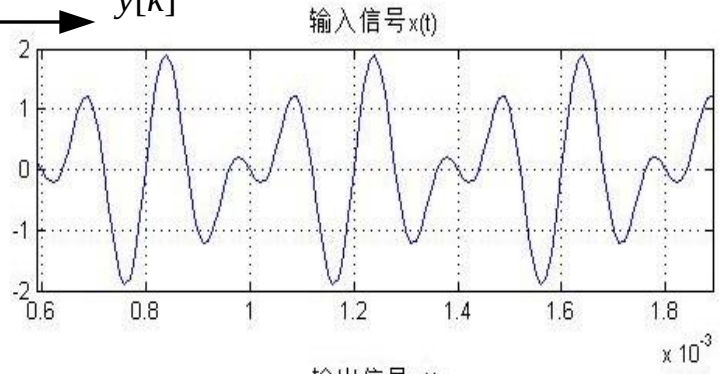
Phase response of second-order real coefficient all pass filter

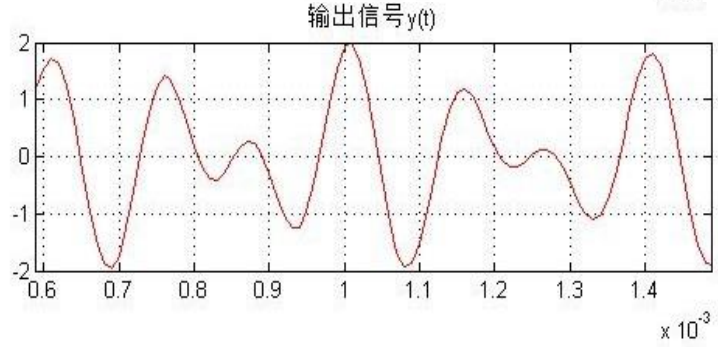
Application of All Pass Digital Filter

The signal passes through a nonlinear phase system







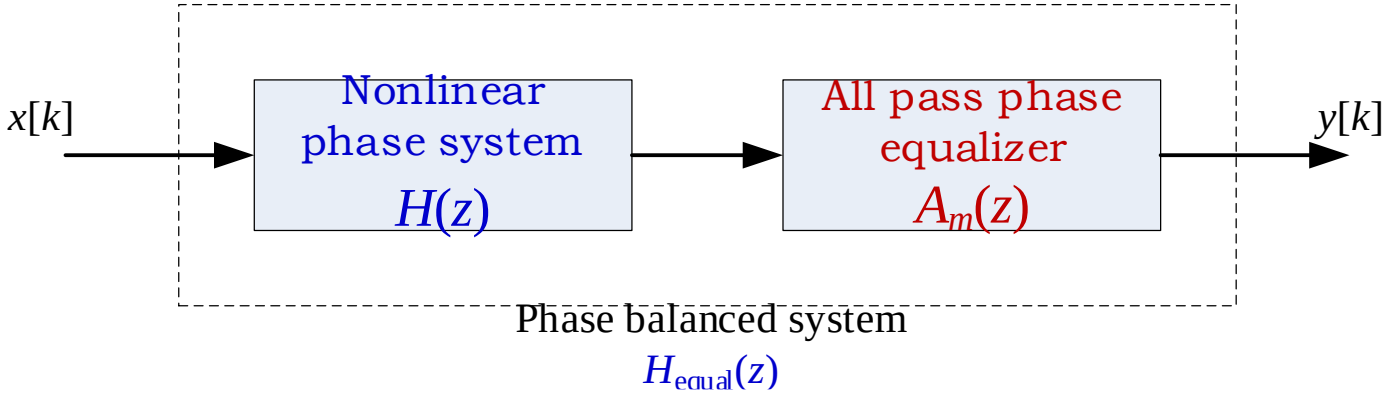


How to solve it?

Compared to $x(t)$, $y(t)$ is significantly distorted

Application of All Pass Digital Filter

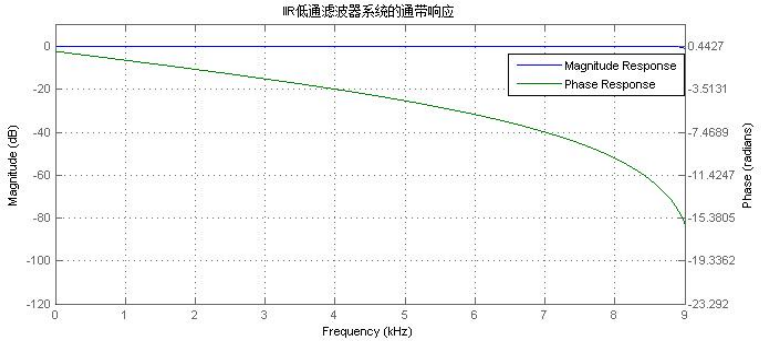
By cascading the all pass system with the original system, the entire system can retain the original amplitude response and meet the required phase characteristics within the required range. The all pass filter that achieves this goal is called a phase equalizer.

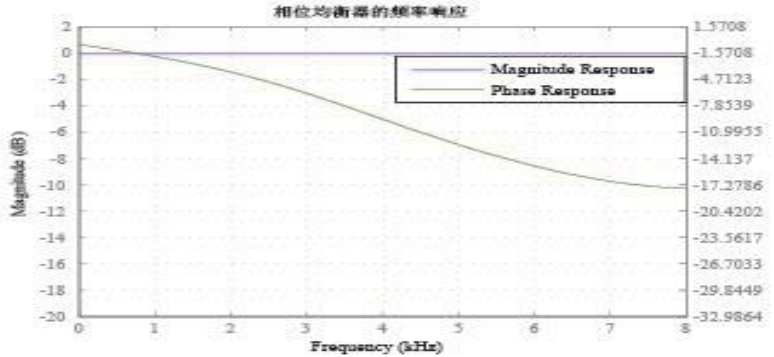


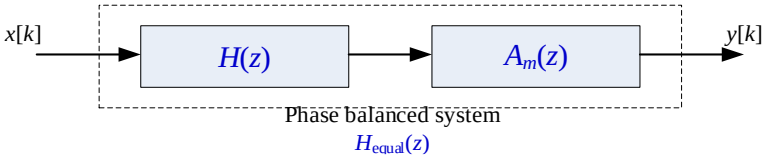
Application of All Pass Digital Filter

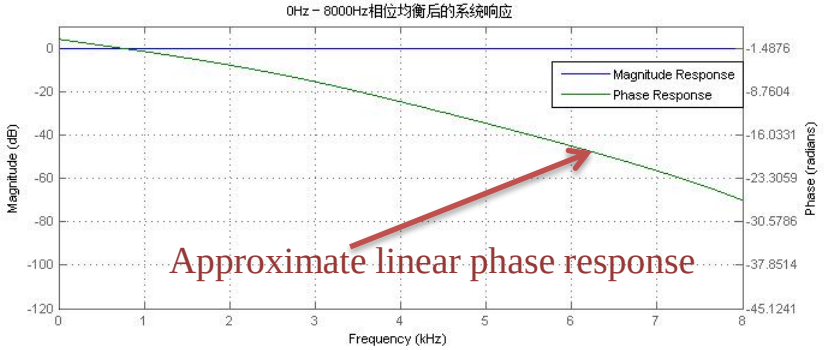
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Design an all pass filter to perform phase equalization on the passband of a nonlinear phase system

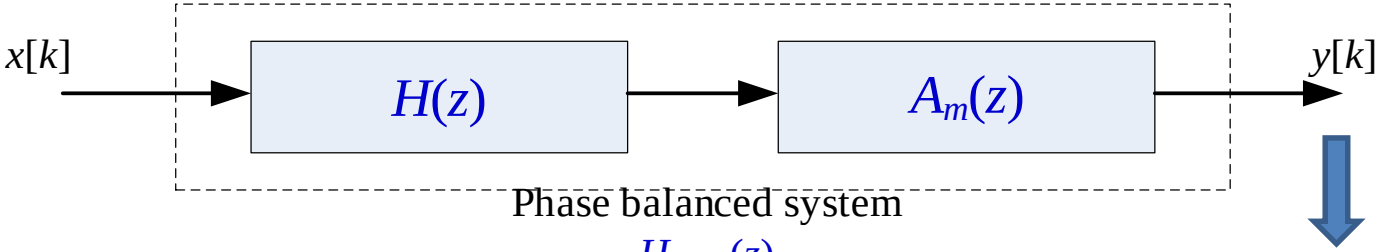


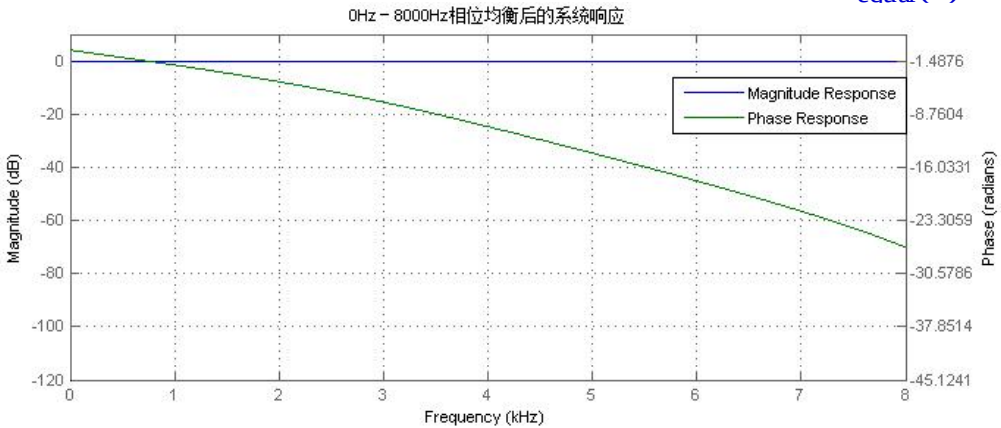




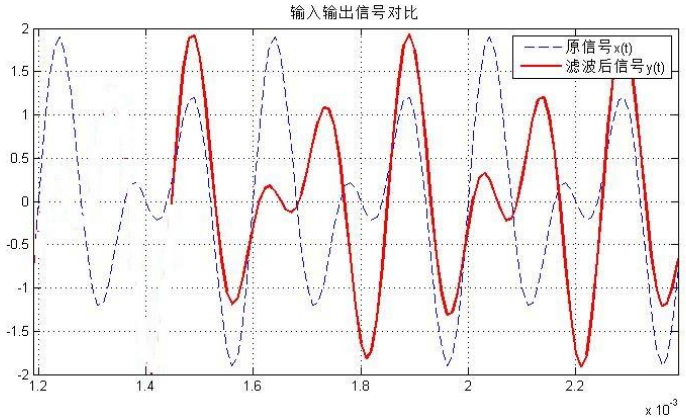


Application of All Pass Digital Filter





Balanced system passband response



Comparison of input and output signals

Minimum phase system

Definition:

The causal system in which the zeros and poles of the system function are both within the unit circle of the z-plane is called the minimum phase system. Recorded as $H_{\min}(z)$.

In the same order system with the same amplitude frequency characteristics, the minimum phase system has the maximum phase and the minimum delay.

Minimum phase system

E.g: Try to determine which causal system $H_1(z)$ or $H_2(z)$ is the minimum phase system, and draw their amplitude and phase responses.

Solution: $H_1(z)=\dfrac{0.4+z^{-1}}{1+0.9z^{-1}}$ $H_2(z)=\dfrac{1+0.4z^{-1}}{1+0.9z^{-1}}$

Two causal systems have the same poles $p_1 = p_2 = -0.9$ Within the unit circle

The zero point of the causal system H_1(z) is $z_1 = -1/0.4 = -2.5$ Outside the unit circle

The zero point of the causal system H_2(z) is $z_2 = -0.4$ Within the unit circle

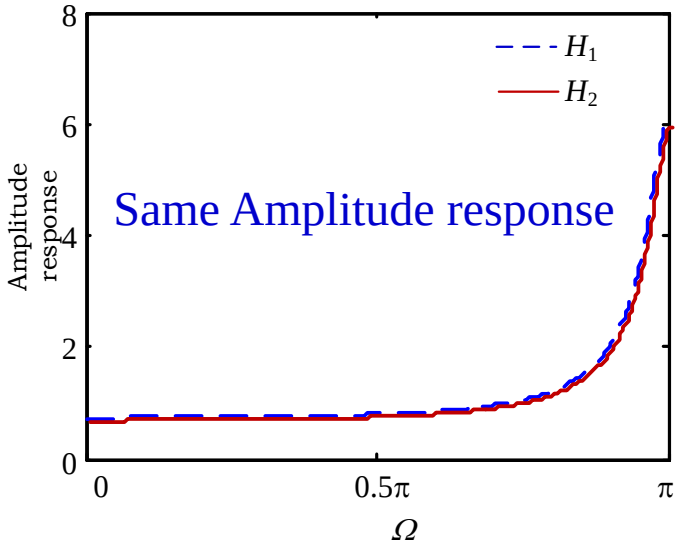
Therefore, $H_2(z)$ is the minimum phase system

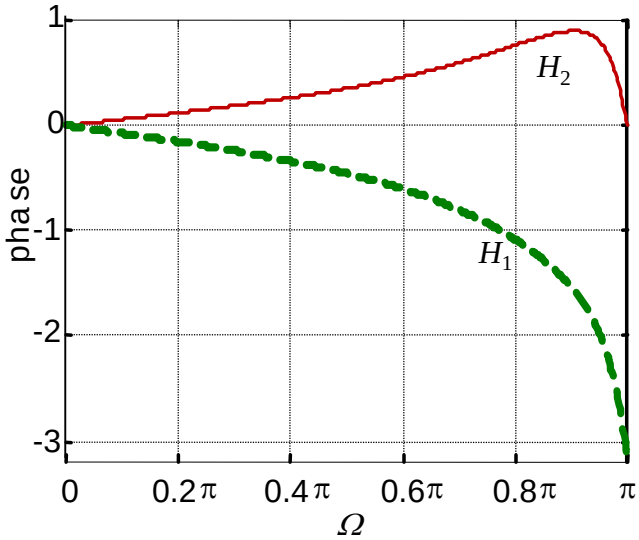
Minimum phase system

E.g: Try to determine which causal system $H_1(z)$ or $H_2(z)$ is the minimum phase system, and draw their amplitude and phase responses.

$$H_1(z)=\frac{0.4+z^{-1}}{1+0.9z^{-1}} \qquad H_2(z)=\frac{1+0.4z^{-1}}{1+0.9z^{-1}}$$

Solution:





The maximum phase of the minimum phase system

Minimum phase system

E.g: The system function $H(z)$ of a certain real coefficient causal stable system is $H(z)=\dfrac{0.4+z^{-1}}{1+0.9z^{-1}}$

Try to find the minimum phase system with the same amplitude response.

Solution: Due to the zero point of the system being $z=-2.5$, it is not the minimum phase system.

$$H(z)=\frac{0.4+z^{-1}}{1+0.9z^{-1}}\cdot\frac{1+0.4z^{-1}}{1+0.4z^{-1}}=\underbrace{\frac{1+0.4z^{-1}}{1+0.9z^{-1}}}_{H_{\min}(z)}\cdot\underbrace{\frac{0.4+z^{-1}}{1+0.4z^{-1}}}_{A(z)}$$

Minimum phase system

Solution:

$$H(z)=\frac{0.4+z^{-1}}{1+0.9z^{-1}}\frac{1+0.4z^{-1}}{1+0.4z^{-1}}=\underbrace{\frac{1+0.4z^{-1}}{1+0.9z^{-1}}}_{H_{\min}(z)}\underbrace{\frac{0.4+z^{-1}}{1+0.4z^{-1}}}_{A(z)}$$

$$\because \left|A(e^{j\Omega})\right|=1 \qquad \therefore \left|H(e^{j\Omega})\right|=\left|H_{\min}(e^{j\Omega})\right|$$

Therefore, the minimum phase system with the same amplitude response as $H(z)$ is

$$H_{\min}(z)=\frac{1+0.4z^{-1}}{1+0.9z^{-1}}$$

The causal system is decomposed into the minimum phase system and the all pass system

E.g: Let $H(z)=\dfrac{(z-0.25)(z-2)}{(z-0.3)(z-0.4)}$ Represented as minimum phase system and all pass system.

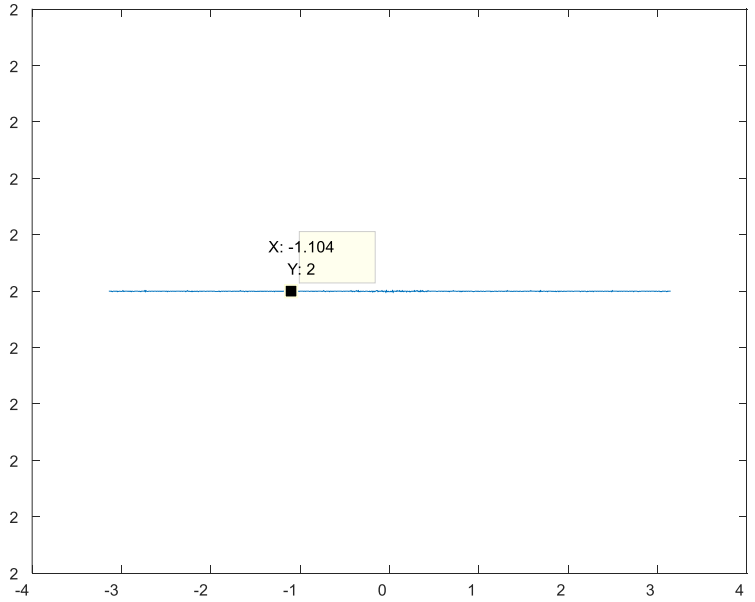
Solution: $H(z)=\dfrac{(z-0.25)(z-2)}{(z-0.3)(z-0.4)}=\dfrac{(z-0.25)(z-2)}{(z-0.3)(z-0.4)}\dfrac{z-1/2}{z-1/2}$

$$=\underbrace{\frac{(z-0.25)(z-1/2)}{(z-0.3)(z-0.4)}}_{H_{\min}(z)}\underbrace{\frac{(z-2)}{(z-1/2)}}_{A(z)}$$

The causal system is decomposed into the minimum phase system and the all pass system

$$A(z)\quad \frac{(z-2)}{(z-1/2)}$$

```
b = [1 -2];
a = [1 -0.5];
W = linspace(-pi,pi,1000);
H = freqz(b,a,W);
plot(W,abs(H));
```



The causal system is decomposed into the minimum phase system and the all pass system

E.g: Find $H(z)=\dfrac{(z-0.25)(z-2)}{(z-0.3)(z-0.4)}$ the minimum phase system with the same amplitude response.

Solution: $$H(z)=\frac{(z-0.25)(z-2)}{(z-0.3)(z-0.4)}=\frac{(z-0.25)(z-2)}{(z-0.3)(z-0.4)}\frac{-2z+1}{-2z+1}$$

$$=\frac{(z-0.25)(-2z+1)}{(z-0.3)(z-0.4)}\frac{(z-2)}{(-2z+1)}$$

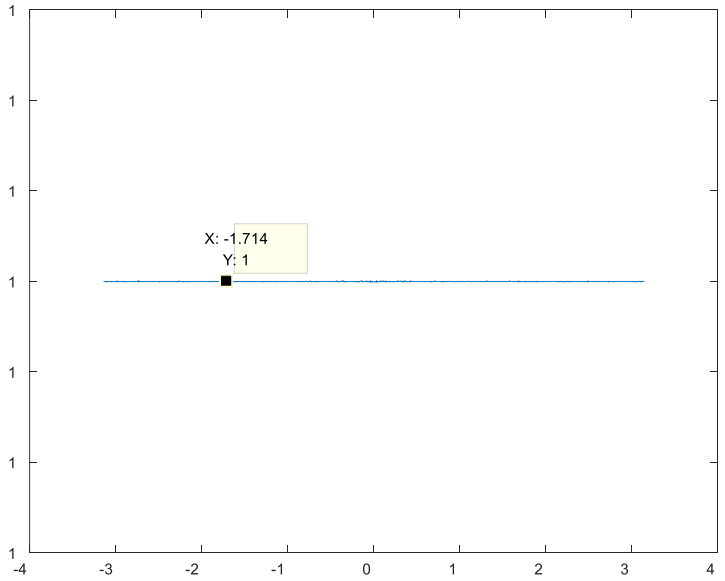
$H_{\min}(z)$ with the same amplitude response as $H(z)$

$A(z)$ with an amplitude response of 1

The causal system is decomposed into the minimum phase system and the all pass system

$$A(z) \quad \frac{(z-2)}{(-2z+1)}$$

```
b = [1 -2];
a = [-2 1];
W = linspace(-pi,pi,1000);
H = freqz(b,a,W);
plot(W,abs(H));
```



The causal system is decomposed into the minimum phase system and the all pass system

E.g: Find $H(z)=\dfrac{(z-0.25)(az+b)}{(z-0.3)(z-0.4)}$ the minimum phase system with the same amplitude response.

Solution: $$H(z)=\frac{(z-0.25)(az+b)}{(z-0.3)(z-0.4)}$$

$$=\frac{(z-0.25)(bz+a)}{(z-0.3)(z-0.4)}\,\frac{(az+b)}{(bz+a)}$$

The causal system is decomposed into the minimum phase system and the all pass system

E.g: Find $H(z)=\dfrac{(1+0.9z^{-1})(1+az^{-1})}{(1+0.3z^{-1})(1+0.4z^{-1})}$ the minimum phase system with the same amplitude response.

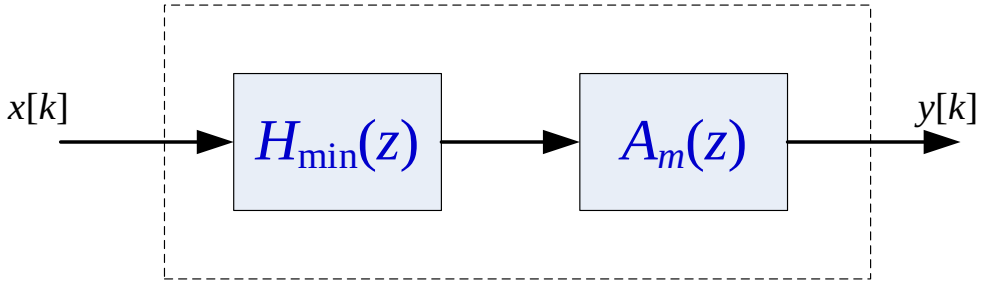
$$H(z)=\frac{(1+0.9z^{-1})(a+z^{-1})}{(1+0.3z^{-1})(1+0.4z^{-1})}\frac{(1+az^{-1})}{(a+z^{-1})}$$

$H_{\min}(z)$ with the same amplitude response as $H(z)$

The causal system is decomposed into the minimum phase system and the all pass system

The $H(z)$ of a causal stable system with real coefficients can be expressed as

$$H(z) = H_{\min}(z) A_m(z)$$



The causal system is decomposed into the minimum phase system and the all pass system

Solution:

$$H(z)= H_{\min}(z)\,A(z) \Longrightarrow |H(e^{j\Omega})| = |H_{\min}(e^{j\Omega})||A(e^{j\Omega})|$$

$$\varphi(e^{j\Omega}) = \varphi_{\min}(e^{j\Omega}) + \varphi_{A}(e^{j\Omega}) \Longrightarrow \varphi_{\min}(e^{j\Omega}) = \varphi(e^{j\Omega}) - \varphi_{A}(e^{j\Omega})$$

$\varphi_{\min}(e^{j\Omega})$: maximum; $\varphi_{A}(e^{j\Omega})$: <0

$$\tau = -\frac{d\varphi(e^{j\Omega})}{d\Omega} \qquad \tau = \tau_{\min} + \tau_{A} \Longrightarrow \tau_{\min} = \tau - \tau_{A}$$

$\tau_{\min}$: minimum; τ_{A}: >0

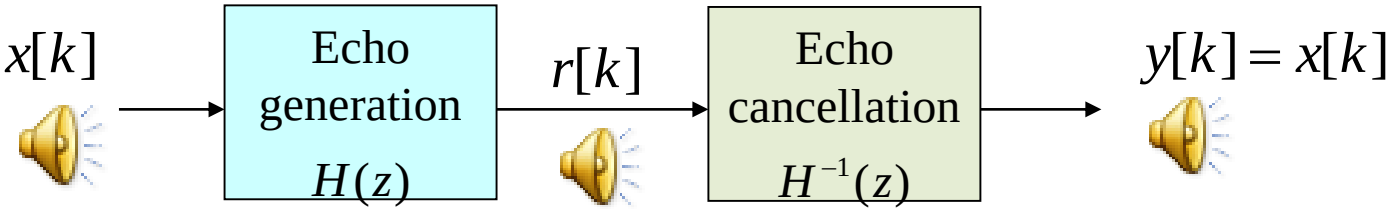
The minimum phase system has the maximum phase and the minimum delay.

The causal system is decomposed into the minimum phase system and the all pass system

Establish a stable system $H(z)=\dfrac{b(z)}{a(z)}$, its inverse system $H^{-1}(z)=\dfrac{a(z)}{b(z)}$

$$H^{-1}(z)H(z)=1$$

$H^{-1}(z)$ can counteract the influence of $H(z)$ and is called an equalizer for $H(z)$.



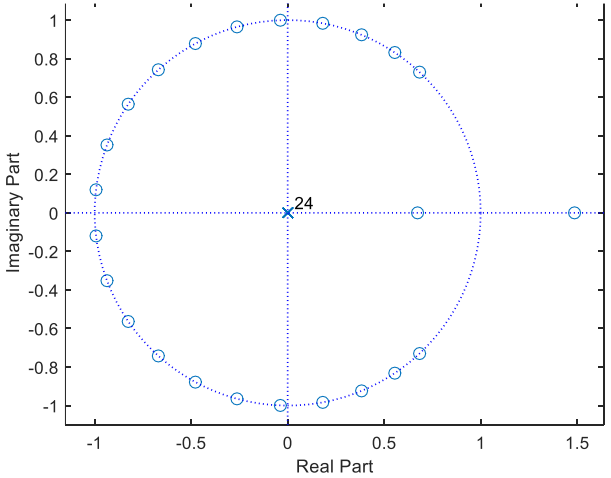
Application of Minimum Phase System

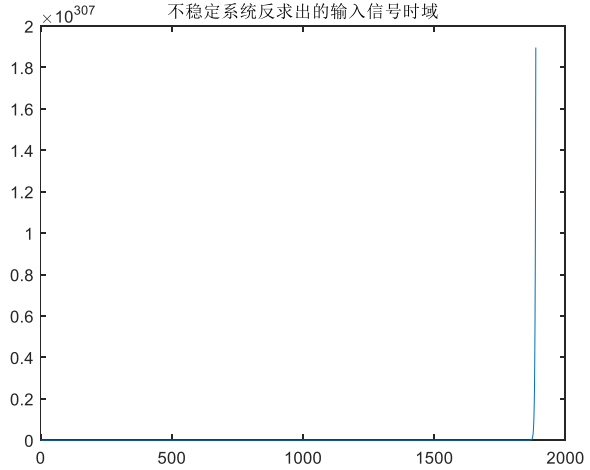
How to solve it?

If there $H(z)=\dfrac{b(z)}{a(z)}$ is a zero point outside the unit circle,

$H^{-1}(z)=\dfrac{a(z)}{b(z)}$ it will be unstable!

When $H(z)$ has a zero point outside the unit circle, let $y[k]$ pass through the system $H^{-1}(z)$ to obtain the following signal:





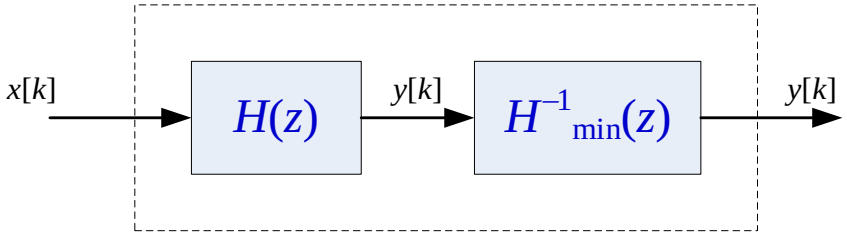
Application of Minimum Phase System

If there $H(z)=\frac{b(z)}{a(z)}$ is a zero point outside the unit circle,

$H^{-1}(z)=\frac{a(z)}{b(z)}$ it will be unstable!

How to solve it?

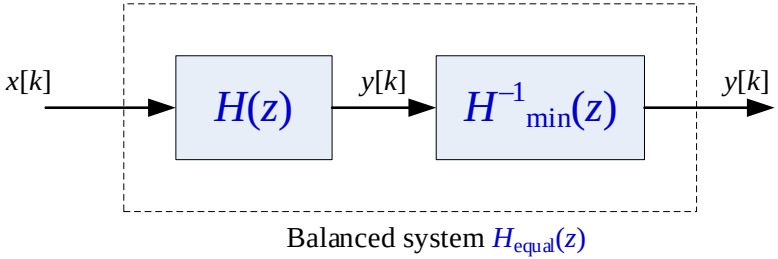
$H(z)$ → Find $H_{\min}(z)$ with the same amplitude response as $H(z)$ → $H_{\min}(z)$ → Obtain a stable inverse system → $H^{-1}_{\min}(z)$



Balanced system $H_{equal}(z)$

Application of Minimum Phase System

Balanced system function

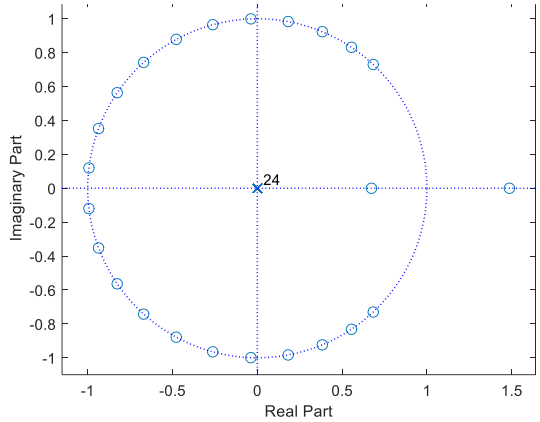


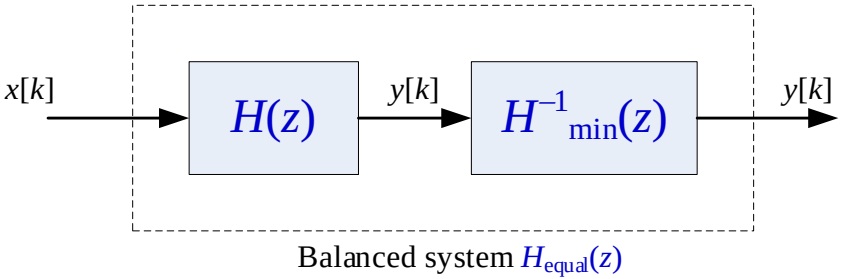
$$H_{\text{equal}}(z) = H^{-1}{}_{\text{min}}(z)H(z) = H^{-1}{}_{\text{min}}(z)H_{\text{min}}(z)A_m(z) = A_m(z)$$

Balanced amplitude response $\left|H_{\text{equal}}(\mathrm{e}^{\mathrm{j}\Omega})\right| = \left|A_m(\mathrm{e}^{\mathrm{j}\Omega})\right| = 1$

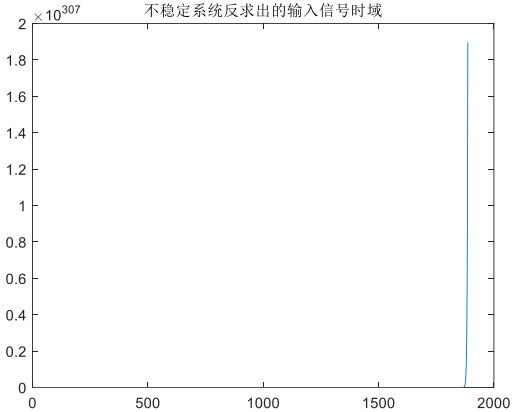
$H^{-1}{}_{\text{min}}(z)$ equalizes the amplitude response of $H(z)$, known as the amplitude equalizer of $H(z)$.

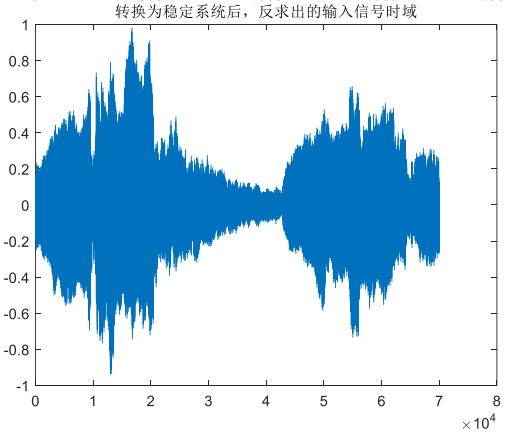
Application of Minimum Phase System





Balanced system $H_{equal}(z)$





Thanks!

The materials referenced in this course are accumulated over years of teaching by the instructor, sourced from various media and exchanges with colleagues and peers. It is difficult to indicate the source of each material one by one. This is to clarify and express gratitude!